SAT UZBEKISTAN

Test creator Xodiyev Lutfulloxon
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1. What is the sum of the 4 binomials listed below?

$$x^2+3, 4x+6,\ 3x^2+1, 3x-1$$

A) $4x^2+7x+9$
B) $4x^2+7x+11$
C) $7x^2+4x+9$
D) $4x^2+4x+11$

2. Which of the following is equivalent to $(3x-5)(-x+7)$?

A. $(3x+5)(x+7)$
B. $(3x-5)(x+7)$
C. $(-3x+5)(x-7)$
D. $(-3x+5)(x+7)$

3. What is the sum of the solutions of the polynomial $f(x)=x^2-7x+12$?

A) -7
B) 3
C) 4
D) 7

4. What are the solutions of the quadratic equation $3x^2+9x-12=0$?

A) $x=1$ and $x=4$
B) $x=-1$ and $x=4$
C) $x=-1$ and $x=-4$
D) $x=1$ and $x=-4$

5. If (x,y) is a solution to the system of equations below, what is a possible value of x?

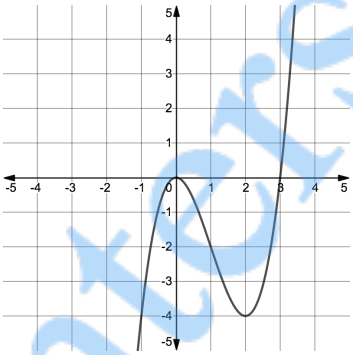
$$y=x^2+6x+6$$
$$y=2x+2$$

A) -2
B) 0
C) 2
D) 4

6. What is the sum of the solutions of the equation $x^2-4x-21=0$?

A) -10
B) -4
C) 3
D) 4

7. The function $f(x)$ is graphed below. Which of the following could define the function $f(x)$?



A. $f(x)=x(x-3)$
B. $f(x)=x(x+3)$
C. $f(x)=x^2(x-3)$
D. $f(x)=x^2(x+3)$

8. In the xy-plane, the parabola with equation $y=(x+3)(x+4)$ intersects the equation $y=20$ at two points. Which of the following is an x-value of a point of intersection?

A) -8
B) -1
C) 3
D) 4

9. In the equation below, a and b are constants. Which of the following could be the value of a?

$$9x^2-16=(ax-b)(ax+b)$$

A) 3
B) 4
C) 9
D) 16

10. In the system of equations below, a and b are constants. For which of the following values of a and b does the system of equations have exactly one real solution?

$$y=6x+2$$
$$y=ax^2+b$$

A) $a=3,\ b=1$
B) $a=3,\ b=3$
C) $a=9,\ b=3$
D) $a=-3,\ b=1$

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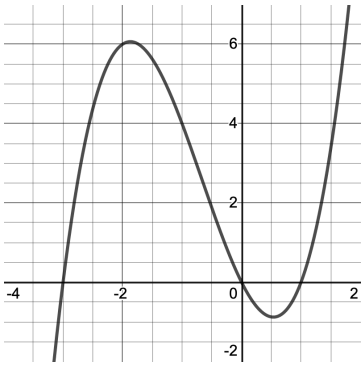
11. What is the sum of the solutions to $(x - 1.2)(x + 5) = 0$?

A) -6.2
B) -3.8
C) 3.8
D) 6.2

12.



The function $f(x)$ is graphed above. Which of the following could define the function $f(x)$?

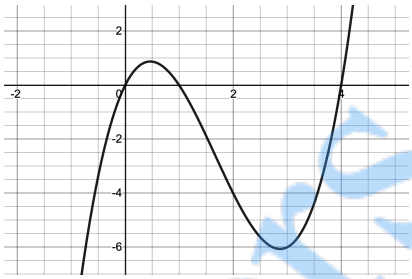
A) $f(x) = (x - 3)(x + 1)$
B) $f(x) = x(x - 3)(x + 1)$
C) $f(x) = (x + 3)(x - 1)$
D) $f(x) = x(x + 3)(x - 1)$

13. In the equation below, j, l, k, and m are constants. If the equation has roots of $-4, 3,$ and -5. Which of the following could be a factor of the equation below?

$$jx^3 + lx^2 - kx - m = 0$$

A) $x - 4$
B) $x - 5$
C) $x - 3$
D) $x + 3$

14.



Which of the following correctly models the graph above?

A) $(x - 1)(x - 4)$
B) $x(x - 1)(x - 4)$
C) $(x + 1)(x + 4)$
D) $x(x + 1)(x + 4)$

15. Which of the following is a solution to the equation below?

$$x^2 + 6x + 3 = 0$$

A) $-3 + \sqrt{6}$
B) $-3 + \sqrt{13}$
C) $3 - \sqrt{6}$
D) $3 - \sqrt{13}$

16. What is the sum of the solutions to the given equation?

$$x^2 - 13x + 40 = 6x - 8$$

A) -13
B) -19
C) 13
D) 19

17. The system of equations below is graphed in the xy-plane. Which of the following is the x-coordinate of an intersection point (x, y) of the system of equations?

$$y = x^2 + 8x + 9$$
$$y = 2x + 3$$

A) $-3 + \sqrt{3}$
B) $3 + 2\sqrt{3}$
C) $-5 - \sqrt{13}$
D) $3 - \sqrt{3}$

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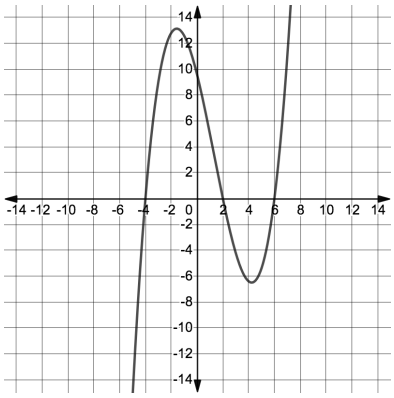
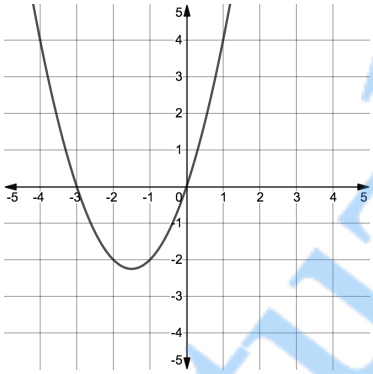
18.



The function $g(x)$ is graphed above. Which of the following could define the function $g(x)$?

A) $g(x) = \frac{1}{5}(x-4)(x+2)(x+6)$
B) $g(x) = -\frac{1}{5}(x-4)(x+2)(x+6)$
C) $g(x) = \frac{1}{5}(x+4)(x-2)(x-6)$
D) $g(x) = -\frac{1}{5}(x+4)(x-2)(x-6)$

19.



The function $f(x)$ is graphed above. Which of the following could define the function $f(x)$?

A) $f(x) = x^2 + 3x$
B) $f(x) = x^2 - 3x$
C) $f(x) = -x^2 + 3x$
D) $f(x) = -x^2 - 3x$

20. Given that $(2x+3)$ and $(x-4)$ are the factors of the quadratic below, what is the value of z?

$$2x^2 + (z-1)x + 2z - 4$$

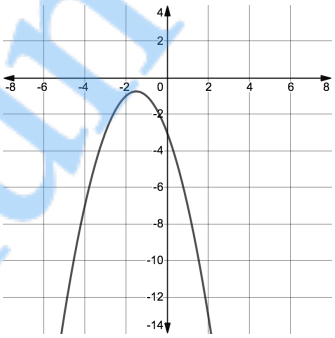
A) -2
B) -4
C) 2
D) 8

21. $$x^3 + 8x^2 - 27x - 28 = 0$$

The polynomial above can be written as $(x+1)(x+7)(x^2-4) = 0$. What are all of the roots of the equation?

A) $-1, -7$
B) $-1, -7, \sqrt{2}$
C) $-2, -1, 2, 7$
D) $-7, -2, -1, 2$

22.



The function $f(x)$ is graphed above. Which of the following could define the function $f(x)$?

A) $f(x) = x^2 - 2x - 3$
B) $f(x) = x^2 - 3x - 3$
C) $f(x) = -x^2 - 3$
D) $f(x) = -x^2 - 3x - 3$

23. The system of equations below is graphed in the xy-plane. Which of the following is the sum of the values of the two x-coordinates of the intersection points (x, y)?

$$y = x^2 + 2x + 1$$
$$y = -3x - 3$$

A) -5
B) -3
C) 3
D) 5

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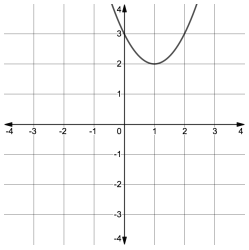
24. Which of the following could be the graph of $f(x) = x^2 - 2x + 3$?

A)



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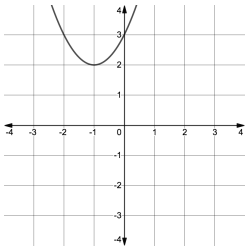

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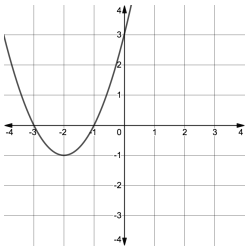

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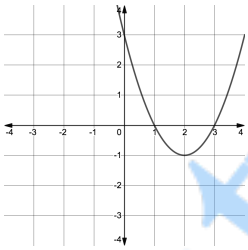


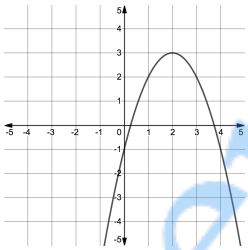
25. The system of equations below is graphed in the xy-plane. If x is not a negative number, what is a possible value of x?

$$y = x^2 + 5x + 8$$
$$y = 8 - 2x$$

A) 0 B) 1 C) 2 D) 3

26. Which of the following could be the graph of $f(x) = -x^2 + 4x - 1$?

A)



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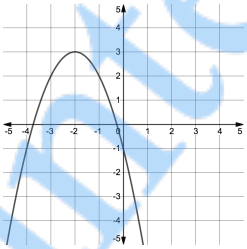

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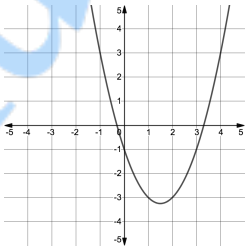
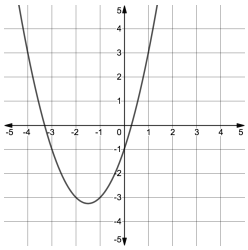


D)



27. Which of the following is a solution to the equation below?

$$x^2 - 4x + 1 = 0$$

A) $2 - \sqrt{6}$
B) $-2 + \sqrt{6}$
C) $2 + \sqrt{3}$
D) $-2 - \sqrt{3}$

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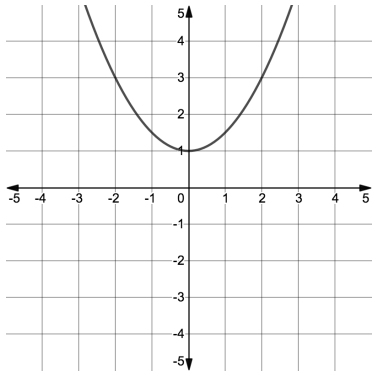
28. The function $f(x)$ is graphed above. Which of the following could define the function $f(x)$?



A) $f(x) = \frac{1}{2}(x-2)(x+2)$
B) $f(x) = (x-2)(x+2)$
C) $f(x) = \frac{1}{2}(x-2)(x+2)+1$
D) $f(x) = \frac{1}{2}(x-2)(x+2)+3$

29. In the quadratic equation below, z is a constant. For what value of z, will the equation have one real solution?

$$zx^2 + 6x = 3$$

A) -3
B) 1
C) 3
D) 6

30. The equation below is graphed in the xy-plane. If a and b are positive constants and $a \neq b$, how many distinct x-intercepts does the graph have?

$$x^2 + ax + bx + ab = 0$$

A) 0
B) 1
C) 2
D) 3

31. What is the solution set for $5x^2 + 6x = 8$?

A) $\{\frac{1}{5}, \frac{1}{2}\}$
B) $\{-\frac{1}{5}, -\frac{1}{2}\}$
C) $\{\frac{4}{5}, 2\}$
D) $\{\frac{4}{5}, -2\}$

32. If (x, y) is a solution to the system of equations below, what is a possible value of $y - x$?

$$y = x^2 + 9x + 8$$
$$y = 11x + 7$$

A) -1
B) 1
C) 17
D) 18

33. What is the sum of the solutions to the given equation?

$$x^2 - 12x + 26 = 2x + 2$$

A) 14
B) 11
C) -11
D) -14

34. How many solutions $(x.y)$ are there to the system of equations below?

$$y = x^2 + 11x + 4$$
$$y = 5x - 5$$

A) 0
B) 1
C) 2
D) 4

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35. $(80x - 42)(15x + 12) = ax^2 + bx + c$

For the equation above, what is the value of $a + b + c$?

A) 1026 B) 1028 C) 1028 D) 1030

36. $(kx + 3)(4x^2 - mx - 3) = 20x^3 - 3x^2 - 24x - 9$

For the equation above, what is the value of km?

A) -15
B) -5
C) 3
D) 15

37. $$ax^3 + bx^2 + cx + d = 0$$

In the function above, a, b, c, and d are all constants. If the equation has roots at -3, 6, and 8, which of the following is a factor of $ax^3 + bx^2 + cx + d$?

A) $x + 1$
B) $x + 3$
C) $x - 3$
D) $x + 6$

38. $$h(x) = x^4 + 2x^3 - 8x^2 - 18x - 9$$

The polynomial above can be written as $(x^2 - 9)(x + 1)^2$. What are all the real roots of the equation?

A) $9, 1$
B) $9, 1, \text{and} - 1$
C) $3, -3, \text{and} - 1$
D) $3, -3, 1, \text{and} - 1$

39. The equation below is graphed in the xy-plane. If a and b are positive constants and $a \neq b$, how many distinct x-intercepts does the graph have?

$$y = (x + a)(x - a)(x + b)^2$$

A) 1
B) 2
C) 3
D) 4

40. Ben is throwing a ball from the top of his building. The ball's height is modeled by the function $H(x) = -x^2 + 10x + 56$, where x is the number of seconds after he throws the ball. How many seconds after throwing the ball does it hit the ground?

A) 4
B) 5
C) 14
D) 56

41. Which of the following is a solution to the equation below?

$$2x^2 + 4x + 1 = 0$$

A) $\frac{-2+\sqrt{2}}{2}$
B) $2\sqrt{2}$
C) $-2 + \sqrt{2}$
D) $\frac{2-\sqrt{2}}{2}$

42. John is launching a rocket. The rocket's height in feet is modeled by the function $H(x) = -x^2 + 30x$, where x is the number of seconds after launch. What is the maximum height of the rocket?

A) 10 feet
B) 15 feet
C) 200 feet
D) 225 feet

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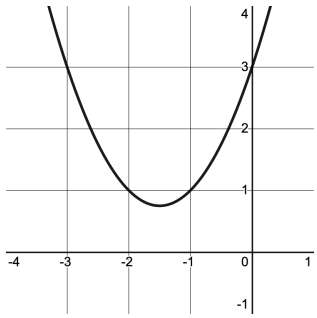

$f(x) = x^2 + bx + 3$

The graph and equation of the function $f(x)$ are shown above. Which of the following is the value of b?

A) 2
B) 3
C) 5
D) 8

44. Dave and Charlie are trying to perform a trick shot. Dave will be on a zip line and his height is represented by the function $f(x) = -3x + 50$, where x represents seconds after starting the zipline. Charlie will be throwing a baseball and the baseball's height is represented by the function $g(x) = -x^2 + 6x + 30$, where x represents the number of seconds after Charlie releases the ball. In order to perform the trick shot, Dave and the baseball must intersect. If Dave starts the zipline at the same time as Charlie releases the ball, which of the following is the most appropriate conclusion about if the trick shot is possible?

A) The trick shot is not possible because the two functions do not intersect.
B) The trick shot is possible and there is only one point of intersection.
C) The trick shot is possible and there are two points of intersection.
D) More information is needed to figure out whether the trick shot is possible.

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